The Regularity Lemma for Stable Graphs and its Applications in Property Testing

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Introduction to Szemerédi's Regularity Lemma

The idea of the SzRL is that all graphs can be partitioned in such a way that most pairs of parts behave in a quasi-random way, a property which is formalized as *regularity*.

Definition

Given $\epsilon > 0$ and a graph G, a pair of (not necessarily disjoint) subsets of vertices $A, B \subseteq G$ is said to be ϵ-*regular* if for all $A' \subseteq A$ and $B' \subseteq B$ such that $|A'| \geq \epsilon|A|$ and $|B'| \geq \epsilon|B|$, we have

$$|d(A', B') - d(A, B)| \leq \epsilon.$$

When most of the pairs satisfy this property, we say that the partition is *regular*.

Definition

Let G be a graph and let $\epsilon > 0$. An *ϵ-regular partition* of G is a partition of its vertex set into k parts $\{A_1, \ldots, A_k\}$ with remainder set B such that:

- $|B| \leq \epsilon|G|$, and may be empty.
- All but at most ϵk^2 of the pairs (A_i, A_j) with $1 \leq i < j \leq k$ are ϵ-regular.

Introduction to Szemerédi's Regularity Lemma

Theorem (Szemerédi's Regularity Lemma, [7])

For every $\epsilon > 0$ and every positive integer m, there exists a positive integer $M = M(\epsilon, m)$ such that every graph with at least m vertices admits an ϵ-regular partition $\{A_1, \dots, A_k\}$ and remainder B with $m \le k \le M$, with $|A_1| = \cdots = |A_k|$.

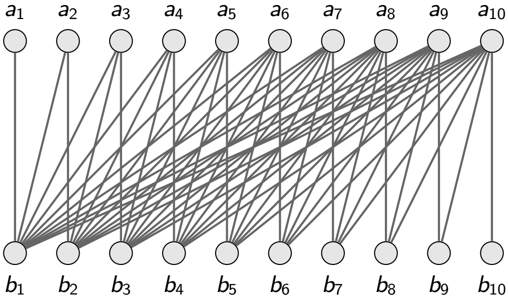
Crucially, the theorem proves that the number of parts k does not grow with the size of the graph, but only depends on ϵ (and m).

However, the lemma has two major drawbacks:

1. The bound on the number of parts M is a *tower-type* function of ϵ, i.e. a power-tower of height polynomial in $1/\epsilon$. This is unavoidable, as shown by Gowers [4].
2. The partition may contain some irregular pairs. This is also unavoidable, and the half-graph H_n is an example.

Stable Graphs and the Stable Regularity Lemma

Stable graphs are those that do not contain a *half-graph* as a bi-induced subgraph.



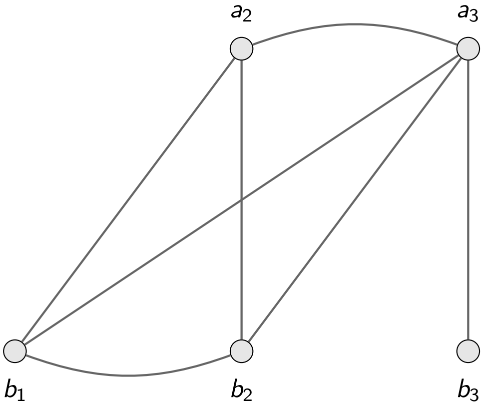
	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8	b_9	b_{10}
a_1	1	0	0	0	0	0	0	0	0	0
a_2	1	1	0	0	0	0	0	0	0	0
a_3	1	1	1	0	0	0	0	0	0	0
a_4	1	1	1	1	0	0	0	0	0	0
a_5	1	1	1	1	1	0	0	0	0	0
a_6	1	1	1	1	1	1	0	0	0	0
a_7	1	1	1	1	1	1	1	0	0	0
a_8	1	1	1	1	1	1	1	1	0	0
a_9	1	1	1	1	1	1	1	1	1	0
a_{10}	1	1	1	1	1	1	1	1	1	1

Stable Graphs and the Stable Regularity Lemma

Definition

Let G be a graph. We say that G has the *k-order property* if there exist two sequences of vertices $\langle a_i \mid i \in \{1, \ldots, k\}\rangle$ and $\langle b_i \mid i \in \{1, \ldots, k\}\rangle$ such that for all $i, j \leq k$, $a_i R b_j$ if and only if $i \geq j$. Otherwise, we say that G has the *non-k-order property* or that G is *k-stable*.

The two sequences need not to be disjoint, and any combination of edges and non-edges may appear within the vertices of each sequence.



	a_2	a_3	b_1	b_2	b_3
a_2	0	1	1	1	0
a_3	1	0	1	1	1
b_1	1	1	0	1	0
b_2	1	1	1	0	0
b_3	0	1	0	0	0

Stable Graphs and the Stable Regularity Lemma

Theorem (Stable Regularity Lemma, [6])

Each k-stable graph G admits an equitable partition of its vertex set with at most $c(k) \cdot (1/\epsilon)^{2^{k+1}}$ parts such that ALL pairs of parts are ϵ-regular and have density either less than ϵ or greater than $1 - \epsilon$.

Notably both limitations of the SzRL are overcome: the bound on the number of parts is polynomial in $1/\epsilon$, and there are no irregular pairs.

Sketch of the Proof of the Stable Regularity Lemma

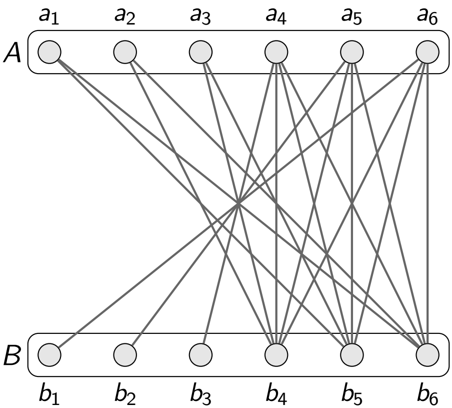
Definition (Good, [6])

Let G be a finite graph with the non-k_*-property. We say that $A \subseteq G$ is *ϵ-good* when for every $b \in G$ the truth value $t = t(b, A) \in \{0, 1\}$ satisfies $|\overline{B}_{A,b}| = |\{a \in A \mid aRb \not\equiv t\}| < \epsilon|A|$.

Definition (Excellent, [6])

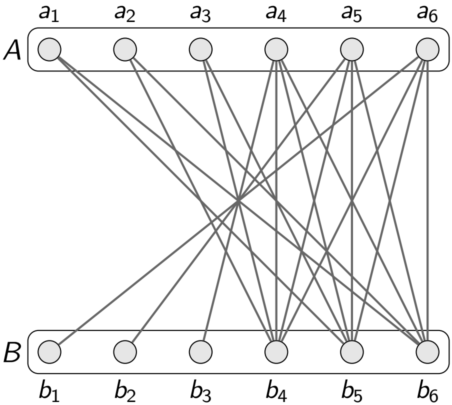
Let G be a finite graph with the non-k_*-property. We say that $A \subseteq G$ is *(ϵ, ζ)-excellent* when A is ϵ-good and, if B is ζ-good, then the truth value $t = t(B, A)$ satisfies $|\{a \in A \mid t(a, B) \neq t(A, B)\}| < \epsilon|A|$. In particular, we say A is *ϵ-excellent* if A is (ϵ, ϵ)-excellent.

Sketch of the Proof of the Stable Regularity Lemma



	b_1	b_2	b_3	b_4	b_5	b_6
a_1	0	0	0	0	1	1
a_2	0	0	0	1	0	1
a_3	0	0	0	1	1	0
a_4	0	0	1	1	1	1
a_5	0	1	0	1	1	1
a_6	1	0	0	1	1	1

Sketch of the Proof of the Stable Regularity Lemma



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a_2	0	0	0	1	0	1
a_3	0	0	0	1	1	0
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a_5	0	1	0	1	1	1
a_6	1	0	0	1	1	1

Do excellent subsets always exist?

Sketch of the Proof of the Stable Regularity Lemma

Definition (k-tree)

A *k-tree* in G is an ordered pair $H = (\overline{c}, \overline{b})$ comprising two (not necessarily disjoint) sequences:

- $\overline{c} = \{c_\eta \in G \mid \eta \in \{0,1\}^{<k}\}$, the set of *nodes* and
- $\overline{b} = \{b_\rho \in G \mid \rho \in \{0,1\}^{k}\}$, the set of *branches*,

satisfying that for all $\eta \in \{0,1\}^{<k}$ and $\rho \in \{0,1\}^{k}$, if for some $\ell \in \{0,1\}$ we have $\eta \frown \langle \ell \rangle \triangleleft \rho$, then $b_\rho R c_\eta \equiv \ell$.

Sketch of the Proof of the Stable Regularity Lemma

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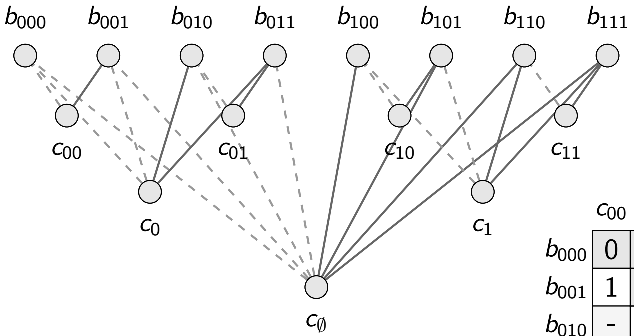
Definition (Tree bound)

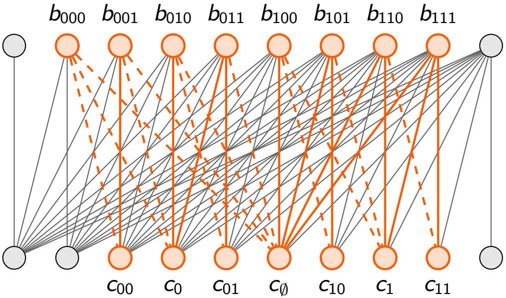
Suppose G is a finite graph. We denote the *tree bound* $k_{**} = k_{**}(G)$ as the minimal positive integer such that there is no k_{**}-tree $H = (\overline{c}, \overline{b})$ in G.

Sketch of the Proof of the Stable Regularity Lemma



	c_{00}	c_0	c_{01}	$c_\emptyset$	c_{10}	c_1	c_{11}
b_{000}	0	0	-	0	-	-	-
b_{001}	1	0	-	0	-	-	-
b_{010}	-	1	0	0	-	-	-
b_{011}	-	1	1	0	-	-	-
b_{100}	-	-	-	1	0	0	-
b_{101}	-	-	-	1	1	0	-
b_{110}	-	-	-	1	-	1	0
b_{111}	-	-	-	1	-	1	1

Sketch of the Proof of the Stable Regularity Lemma



			c_{00}	c_0	c_{01}	$c_\emptyset$	c_{10}	c_1	c_{11}	
	1	0	0	0	0	0	0	0	0	0
b_{000}	1	1	0	0	0	0	0	0	0	0
b_{001}	1	1	1	0	0	0	0	0	0	0
b_{010}	1	1	1	1	0	0	0	0	0	0
b_{011}	1	1	1	1	1	0	0	0	0	0
b_{100}	1	1	1	1	1	1	0	0	0	0
b_{101}	1	1	1	1	1	1	1	0	0	0
b_{110}	1	1	1	1	1	1	1	1	0	0
b_{111}	1	1	1	1	1	1	1	1	1	0
	1	1	1	1	1	1	1	1	1	1

Sketch of the Proof of the Stable Regularity Lemma

Theorem (Lemma 6.7.9 in [5])

If a graph G has the $2^{k_{**}}$-order property, then the tree bound of G is at least $k_{**}+1$.

Sketch of the Proof of the Stable Regularity Lemma

Theorem (Lemma 6.7.9 in [5])

If a graph G has the $2^{k_{**}}$-order property, then the tree bound of G is at least $k_{**}+1$.
On the other hand, if a graph G has tree bound at least $k_{**} = 2^{k_*+1}-1$, then it has the k_*-order property.

Sketch of the Proof of the Stable Regularity Lemma

Lemma (Existence of excellent subsets, [6])

Let G be a finite graph with the non-k_*-order property. Let $\zeta \leq \frac{1}{2^{k_{**}}}$, $\epsilon \in (0, \frac{1}{2})$. Then, for every $A \subseteq G$ with $|A| \geq \frac{1}{\epsilon^{k_{**}}}$ there exists an (ϵ, ζ)-excellent subset $A' \subseteq A$ such that $|A'| \geq \epsilon^{k_{**}-1}|A|$.

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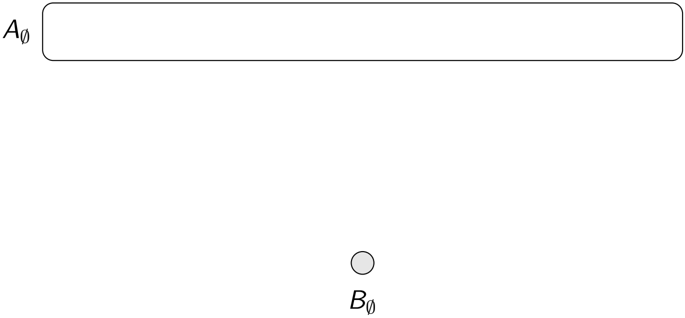
TO DO

Sketch of the Proof of the Stable Regularity Lemma

$A_\emptyset$

Sketch of the Proof of the Stable Regularity Lemma



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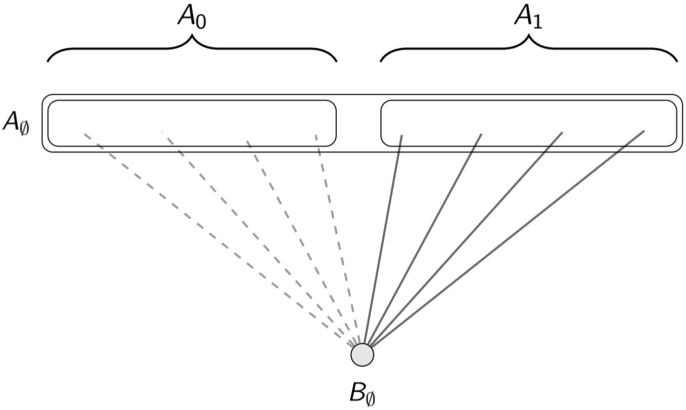

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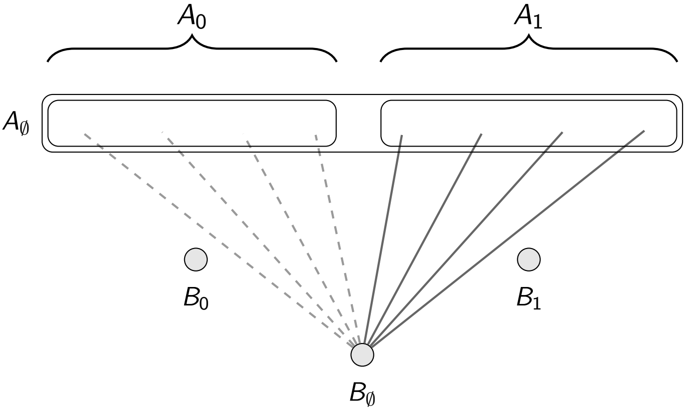

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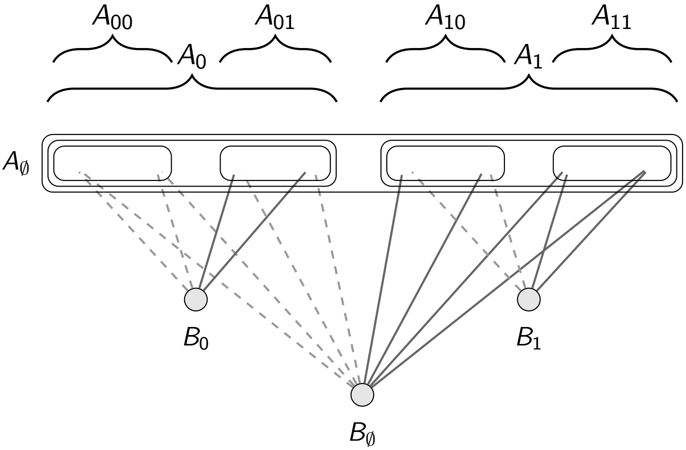

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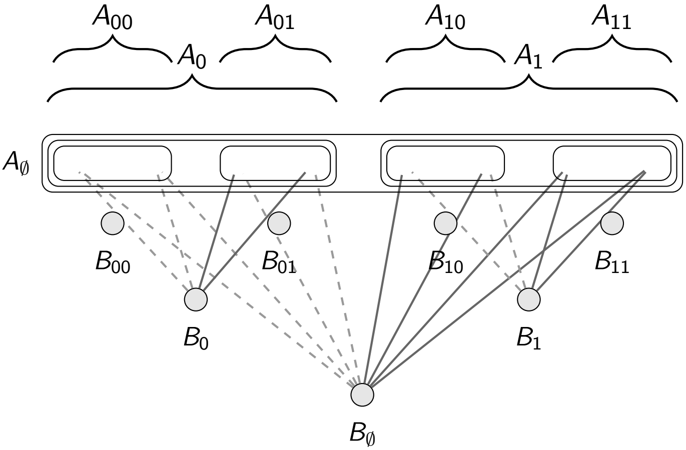

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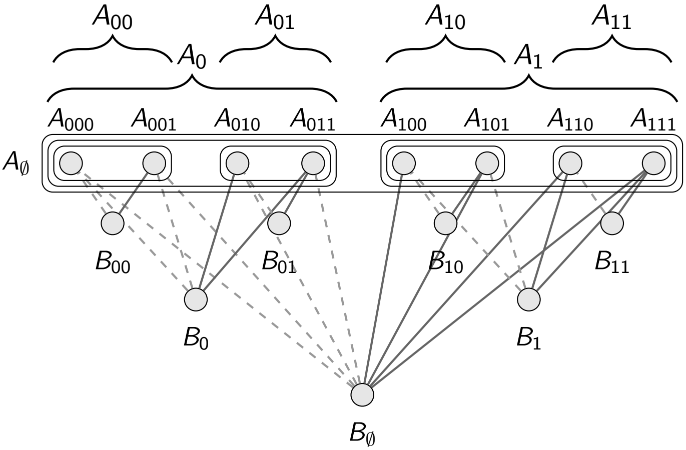

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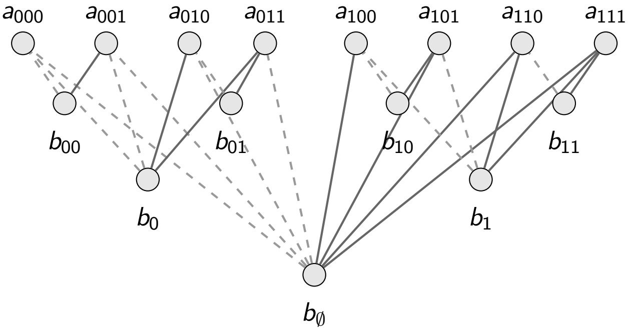

Sketch of the Proof of the Stable Regularity Lemma

Lemma (Existence of excellent subsets, [6])

Let G be a finite graph with the non-k_*-order property. Let $\zeta < \frac{1}{2^{k_{**}}}$, $\epsilon \in (0, \frac{1}{2})$. Let $\langle m_\ell \mid \ell \in \{0, \ldots, k_{**}\}\rangle$ be a decreasing sequence of natural numbers such that $\epsilon m_\ell \geq m_{\ell+1}$ for all $\ell \in \{0, \ldots, k_{**} - 1\}$ and $m_{k_{**}} \geq 1$. Then, for every $A \subseteq G$ with $|A| \geq m_0$ there exists $(\frac{m_{\ell+1}}{m_\ell}, \zeta)$-excellent subset $A' \subseteq A$ such that $|A'| = m_\ell$ for some $\ell \in \{0, \ldots, k_{**} - 1\}$.

Sketch of the Proof of the Stable Regularity Lemma

Lemma

Let G be a finite graph with the non-k_*-order property. Let $\epsilon \in (0, \frac{1}{2})$ and $\epsilon' \leq \frac{1}{2^{k_{**}}}$. Let $A \subseteq G$ such that $|A| = n \geq m_0$. Let $\langle m_\ell \mid \ell \in \{0, \ldots, k_{**}\}\rangle$ be a decreasing sequence of natural numbers such that $\epsilon m_\ell \geq m_{\ell+1}$ for all $\ell \in \{0, \ldots, k_{**} - 1\}$ and $m_{k_{**}} \geq 1$. Denote $m_* := m_0$ and $m_{**} := m_{k_{**}}$. Then, there is a partition $\overline{A} = \langle A_j \mid j \in \{1, \ldots, j(*)\}\rangle$ with remainder $B = A \setminus \bigcup_{j<j(*)} A_j$ such that:

1. *For all $j \in \{1, \ldots, j(*)\}$, $|A_j| \in \langle m_\ell \mid \ell \in \{0, \ldots, k_{**} - 1\}\rangle$.*
2. *For all $i \neq j \in \{1, \ldots, j(*)\}$, $A_i \cap A_j = \emptyset$.*
3. *For all $j \in \{1, \ldots, j(*)\}$, A_j is (ϵ, ϵ')-excellent.*
4. *$|B| < m_*$.*

Sketch of the Proof of the Stable Regularity Lemma

Question: Are subsets of excellent sets still excellent?

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Answer: With high probability, yes, but with a small loss in excellence.

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Lemma

Let G be a finite graph with the non-k_*-order property. Then, for every $\epsilon \in (0, \frac{1}{2})$, $\zeta \in (0, \frac{1}{2} - \epsilon)$, $\xi \in (0, 1)$ and $m \geq \frac{1}{\zeta^2}(k_* \log \frac{1}{\zeta^2} k_* - \log \xi)$, if $A \subseteq G$ is an ϵ-good subset of size $n \geq m$, then a subset $A' \subseteq A$ of size m, sampled uniformly at random, is $(\epsilon + \zeta)$-good with probability $1 - \xi$.

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Answer: With high probability, yes, but with a small loss in excellence.

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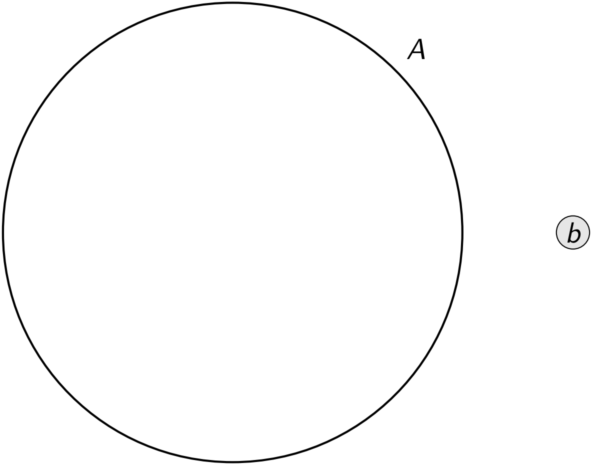
Lemma

Let G be a finite graph with the non-k_*-order property. Then, for every $\zeta \in \{0, \frac{1}{2}\}$ and $\zeta' < \zeta$, there is $\epsilon_1 = \epsilon_1(\zeta, \zeta')$ such that for every $\epsilon < \epsilon' \leq \epsilon_1$, if

- *$A \subseteq G$ is (ϵ, ϵ')-excellent and*
- *$A' \subseteq A$ is $(\epsilon + \zeta')$-good,*

then, A' is $(\epsilon + \zeta, \epsilon')$-excellent.

Sketch of the Proof of the Stable Regularity Lemma



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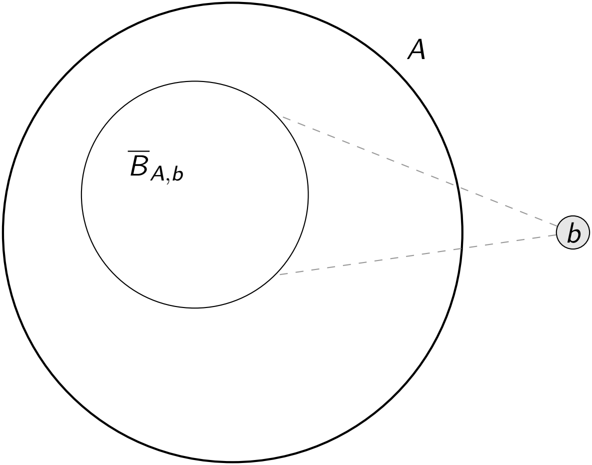

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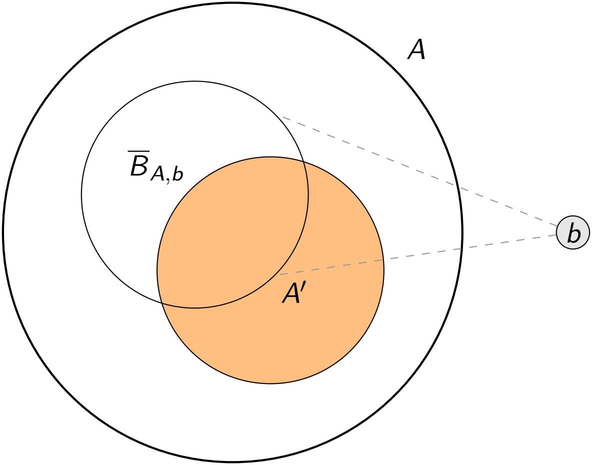

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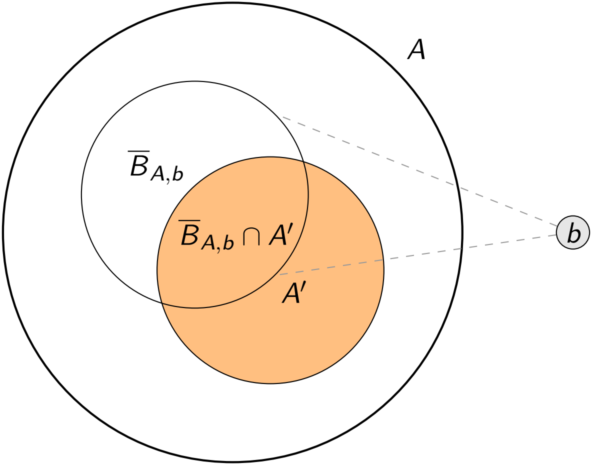

Sketch of the Proof of the Stable Regularity Lemma

The probability of failure is small for each b:

$$\mathbb{P}(X_{A,b,A'}) \leq \mathbb{P}\left(|\overline{B}_{A,b} \cap A'| \geq \left(\frac{|\overline{B}_{A,b}|}{|A|} + \zeta\right)|A'|\right) \leq e^{-2\zeta^2|A'|}$$

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Too many b's?

Sketch of the Proof of the Stable Regularity Lemma

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Too many b's?

Lemma

Let G be a graph with the (disjoint) non-k-order property. Then, for any finite non-trivial $A' \subseteq G$,

$$|\{\overline{B}_{A',b} \mid b \in G\}| \leq |A'|^k.$$

Sketch of the Proof of the Stable Regularity Lemma

Given the random variable

$$X = \sum_{\text{bad } \overline{B}_{A,b}} \mathbb{1}_{X_{A,b,A'}},$$

Sketch of the Proof of the Stable Regularity Lemma

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we can compute its expectation

$$\mathbb{E}[X] = \sum_{\text{bad } \overline{B}_{A,b}} \mathbb{E}[\mathbb{1}_{X_{A,b,A'}}] = \sum_{\text{bad } \overline{B}_{A,b}} \mathbb{P}(X_{A,b,A'}) \leq \sum_{\text{bad } \overline{B}_{A,b}} e^{-2\zeta^2 m},$$

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and by the First Moment Method

$$\mathbb{P}(X \geq 1) \leq \mathbb{E}[X] \leq m^{k_*} \cdot e^{-2\zeta^2 m} \leq \xi.$$

Sketch of the Proof of the Stable Regularity Lemma

Lemma

Let G be a finite graph with the non-k_*-order property. Then, for every $\zeta \in \{0, \frac{1}{2}\}$ and $\zeta' < \zeta$, there is $\epsilon_1 = \epsilon_1(\zeta, \zeta')$ such that for every $\epsilon < \epsilon' \leq \epsilon_1$, if

- *$A \subseteq G$ is (ϵ, ϵ')-excellent and*
- *$A' \subseteq A$ is $(\epsilon + \zeta')$-good,*

then, A' is $(\epsilon + \zeta, \epsilon')$-excellent.

Proof idea: given a good set B, the number of exceptional edges between A and B, and thus between A' and B, is small, since A is (ϵ, ϵ')-excellent. Also, since B is good, each vertex in A' either has few or many exceptional edges with B. These two facts together imply that the number of vertices in A' that do not behave as the majority is small.

Sketch of the Proof of the Stable Regularity Lemma

Now, we know that random subsets of excellent sets are (slightly less) excellent with high probability. We can use this fact with a simple Union Bound argument to prove the following result.

Lemma

Let G be a finite graph with the non-k_*-order property. Then, for all $\zeta \in (0, \frac{1}{2})$, $\zeta' < \zeta$, $r \geq 1$ and for all $\epsilon < \epsilon'$ small enough ($\leq \epsilon_1(\zeta, \zeta')$ from the previous point) there exists $N = N(k_*, \zeta', r)$ such that: if $|A| = n > N$, r divides n and A is (ϵ, ϵ')-excellent, then there exists a partition into r disjoint pieces of equal size, each of which is $(\epsilon + \zeta, \epsilon')$-excellent.

Sketch of the Proof of the Stable Regularity Lemma

Applying this result to the previously obtained uneven partition, and carefully choosing some parameters, we obtain an even partition into excellent parts.

Lemma

Let G be a finite graph with the non-k_*-order property. Let ϵ' and ϵ be two real numbers such that $\epsilon' \leq \min\{\frac{1}{5}, \frac{1}{2^{k_{**}}}\}$ and $\epsilon \leq \frac{1}{4k}\epsilon'$ for some $k > 1$. Also, let m_*, m_{**} and q be natural numbers such that $q \geq \lceil \frac{1}{\epsilon} \rceil$, $m_{**} > \frac{N(k, k_*, \epsilon', \frac{m_*}{m_{**}})}{q}$ and $m_* := q^{k_{**}} m_{**}$. Then, for any $A \subseteq G$ with $|A| = n \geq m_*$ there exists a partition $\overline{A} = \langle A_i \mid i \in \{1, \ldots, i(*)\} \rangle$ with remainder $B = A \setminus \bigcup \overline{A}$ such that:

1. $i(*) \leq \frac{n}{m_{**}}$.
2. *For all* $i \in \{1, \ldots, i(*)\}$, $|A_i| = m_{**}$.
3. *For all* $i \in \{1, \ldots, i(*)\}$, A_i *is* $(\frac{\epsilon'}{k}, \epsilon')$*-excellent.*
4. $|B| < m_*$.

Sketch of the Proof of the Stable Regularity Lemma

This can be addressed by uniformly distributing the vertices of the remainder among the parts. This, again, slightly reduces the excellence of each part.

Lemma

Under the same condition of the previous lemma, we can get a partition $\overline{A} = \langle A_i \mid i \in \{1, \ldots, i(*)\} \rangle$ *with no remainder, such that:*

1. *For all* $i, j \in \{1, \ldots, i(*)\}$, $||A_i| - |A_j|| \leq 1$.
2. *For all* $i, j \in \{1, \ldots, i(*)\}$, $A_i \cap A_j = \emptyset$.
3. *For all* $i \in \{1, \ldots, i(*)\}$, A_i *is* (ϵ'', ϵ')*-excellent, for some* ϵ'' *satisfying*

$$0 < \epsilon'' \leq \frac{\frac{\epsilon'}{k} m_{**} + \left\lceil \frac{m_*}{i(*)} \right\rceil}{m_{**} + \left\lceil \frac{m_*}{i(*)} \right\rceil}.$$

4. $A = \bigcup \overline{A}$.

Sketch of the Proof of the Stable Regularity Lemma

By carefully choosing hyperparameters, we can rephrase the result to only depend on the stability constant k_*, the desired excellence ϵ and the minimum partition size m.

Theorem

Let k_* be given. Then, for all $\epsilon \leq \min\{\frac{1}{5}, \frac{1}{2^{k_{**}}}\}$ and $m > 1$, there is $M = M(\epsilon, m, k_*)$ and $N = N(\epsilon, m, k_*)$ such that, for every finite graph G with the non-k_*-order property, and every $A \subseteq G$ with $|A| \geq N$, there exists a partition $\overline{A} = \langle A_i \mid i \in \{1, \ldots, i(*)\}\rangle$ of A, such that:

1. *The number of parts is bounded by* $m \leq i(*) \leq M := \max\{\lceil \frac{12}{\epsilon} \rceil^{k_{**}+1}, 4m\}$.
2. *For all* $i, j \in \{1, \ldots, i(*)\}$, $||A_i| - |A_j|| \leq 1$.
3. *For all* $i \in \{1, \ldots, i(*)\}$, A_i *is* ϵ*-excellent.*

Sketch of the Proof of the Stable Regularity Lemma

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Lemma

For any $\epsilon_0, \epsilon \in (0, \frac{1}{2})$ such that $\epsilon_0 \leq \frac{\epsilon^2}{2}$, and for any two ϵ_0-excellent sets A, B, we have that:

1. *(A, B) is ϵ-regular.*
2. *If $A' \in [A]^{\geq \epsilon |A|}$ and $B' \in [B]^{\geq \epsilon |B|}$, then $d(A', B') < \epsilon$ or $d(A', B') \geq 1 - \epsilon$.*

Sketch of the Proof of the Stable Regularity Lemma

Theorem (Stable Regularity Lemma, [6])

For every $k_* \in \mathbb{N}$ and $\epsilon \in (0, \frac{1}{2})$ and $m > 1$, there exist $N = N(\epsilon, m, k_*)$ and $M = M(\epsilon, m, k_*)$ such that, for every finite graph G with the non-k_*-order property, and every $A \subseteq G$ with $|A| \geq N$, the following holds. There exists $m < \ell < M$ and a partition $\overline{A} = \langle A_i \mid i \in \{1, \dots, \ell\} \rangle$ of A such that each A_i is $\frac{\epsilon^2}{2}$-excellent and, for every (not necessarily distinct) $i, j \in \{1, \dots, \ell\}$,

1. $||A_i| - |A_j|| \leq 1$, *i.e. the partition is equitable.*
2. (A_i, A_j) *is* ϵ*-regular, and moreover if* $B_i \in [A_i]^{\geq \epsilon |A_i|}$ *and* $B_j \in [A_j]^{\geq \epsilon |A_j|}$, *then either* $d(B_i, B_j) < \epsilon$ *or* $d(B_i, B_j) \geq 1 - \epsilon$.
3. *If* $\epsilon \leq \min\{\frac{1}{5}, \frac{1}{2^{k_{**}}}\}$, *then* $M \leq \max\{\left\lceil \frac{12}{\epsilon} \right\rceil^{k_{**}+1}, 4m\}$.

Sketch of the Proof of the Stable Regularity Lemma

The fact that the pairs are either almost empty or almost full is not a core property of stable graphs, and is in fact proven in [3] to be satisfied by a larger class of graphs, the graphs with bounded *VC-dimension*. However, such graphs still require irregular pairs, something that stable graphs completely avoid.

Property Testing of H-freeness in Stable Graphs

Definition

We say that a graph G is ϵ-*far* from satisfying a graph property $\mathcal{P}$ if no adding or removing of up to $\epsilon\binom{|G|}{2}$ edges in G results in the graph satisfying the property.

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Definition

An ϵ*-test* $\mathcal{A}$ deciding a graphs property $\mathcal{P}$ with query complexity $q(n)$ is a randomized algorithm that, on input graph G of size n, satisfies the following.

1. If $G \in \mathcal{P}$, then $\mathbb{P}(\mathcal{A} \text{ accepts } G) \geq \frac{2}{3}$.
2. If G is ϵ-far from satisfying $\mathcal{P}$, then $\mathbb{P}(\mathcal{A} \text{ rejects } G) \geq \frac{2}{3}$.

The query complexity $q(n)$ is the maximum number of queries the algorithm can make, and (in our case) a query discerns whether a desired pair of vertices in the input graph G is adjacent or not.

Property Testing of H-freeness in Stable Graphs

Definition

We say that a property $\mathcal{P}$ is *testable* if there exists an ϵ-test deciding $\mathcal{P}$ with a constant query-complexity with respect to the size of the input graph, that is, it only depends on the parameter ϵ.

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Alon et al. [1] proved that H-freeness is testable for every H. Later, Alon and Shapira [2] generalize this result to any hereditary property.

Theorem (Alon & Shapira Theorem)

Every hereditary graph property is testable (with one-sided error).

Property Testing of H-freeness in Stable Graphs

The main theoretical result we require to build our tester is that if a graph is *far* from being H-free, then it contains many copies of H.

Definition

A graph H is γ*-unavoidable* in a graph G if no adding or removing of up to $\gamma\binom{|G|}{2}$ edges in G results in H not appearing as an induced subgraph of G.

Definition

A graph H is η*-abundant* in a graph G if G contains at least $\eta|G|^{|H|}$ induced copies of H.

Property Testing of H-freeness in Stable Graphs

Lemma (Unavoidability implies abundance, [1])

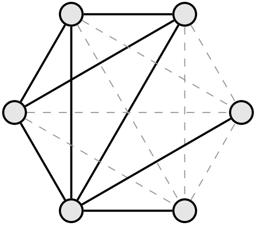
For every real number $\delta \in (0,1)$ and integer $\ell > 0$ there exist $\epsilon = \epsilon(\delta, \ell)$ and $\eta = \eta(\delta, \ell)$ satisfying the following property:
Let H be a graph with vertices $v_1, \ldots, v_\ell$ and let $V_1, \ldots, V_\ell$ be an ℓ-tuple of (not necessarily disjoint) sets of vertices of a graph G such that for every $1 \leq i < i' \leq \ell$, the pair $(V_i, V_{i'})$ is ϵ-regular, with density at least δ if $v_i v_{i'}$ is an edge of H, and at most $1 - \delta$ if $v_i v_{i'}$ is not an edge of H.
Then, at least $\eta \prod_{i=1}^{\ell} |V_i|$ of ℓ-tuples $w_1 \in V_1, \ldots, w_\ell \in V_\ell$ span induced copies of H where w_i plays the role of v_i.

Property Testing of H-freeness in Stable Graphs

Theorem

For every k_*, γ, ℓ there is a $\eta(k_*, \gamma, \ell)$ such that if H is a graph with ℓ vertices, G has the non-k_*-order property and H is γ-unavoidable in G, then H is η-abundant in G.

Property Testing of H-freeness in Stable Graphs



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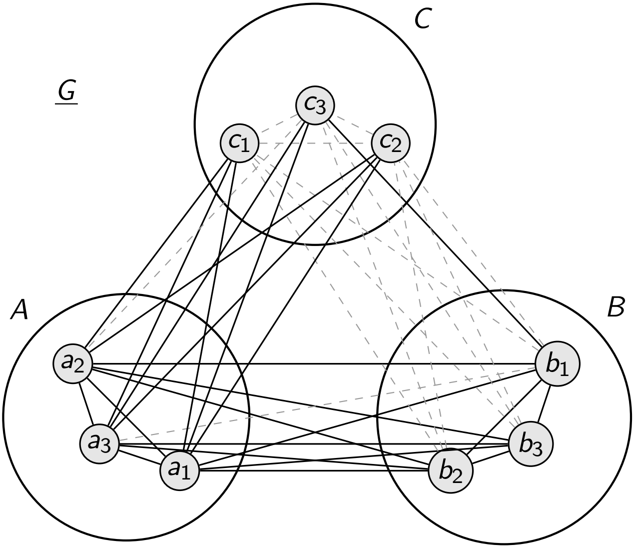
Property Testing of H-freeness in Stable Graphs



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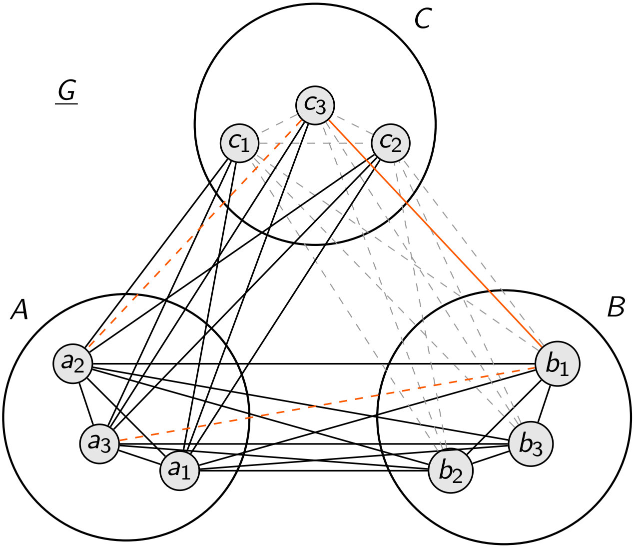

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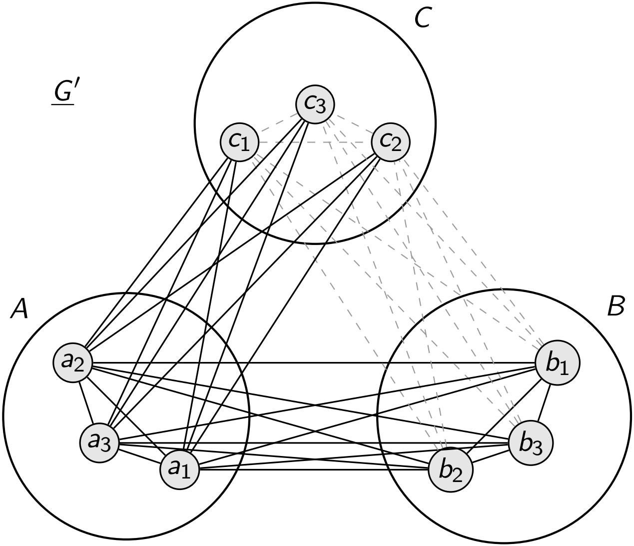

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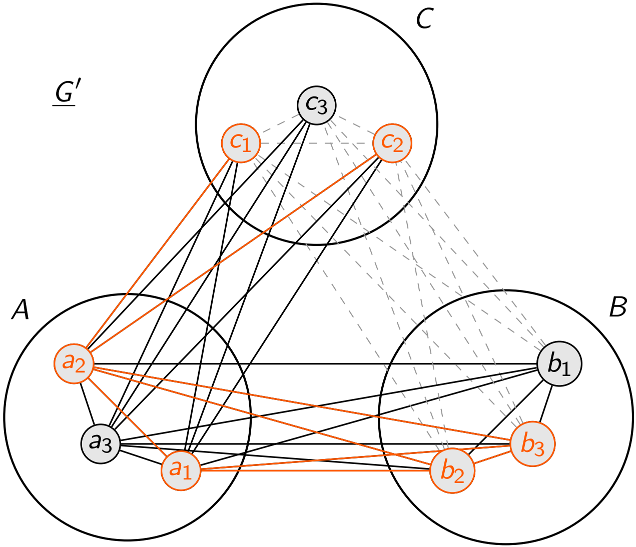

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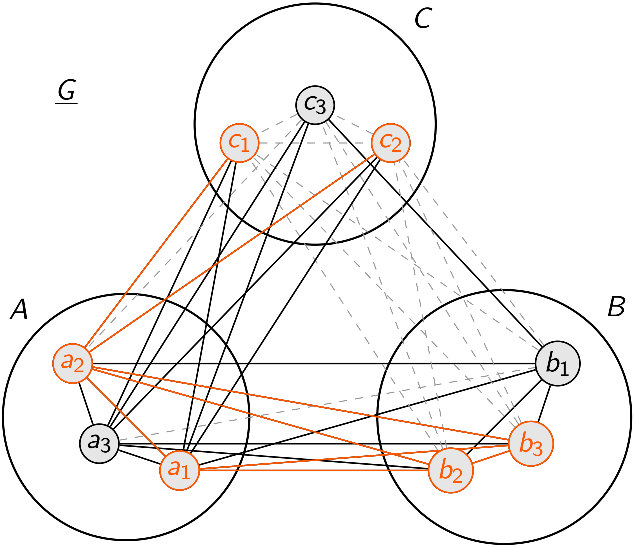

Property Testing of H-freeness in Stable Graphs

Algorithm 1 ϵ-test $\mathcal{A}$ for deciding H-freeness

Require: a graph H of size ℓ, a natural number k_* and a real number $\epsilon > 0$.
Require: an oracle $\mathcal{O}$ accepting queries of whether two vertices of a graph G are adjacent. The graph G has the non-k_*-order property, and only its size n is known.

$t \leftarrow \ell \log(\frac{2}{3}) / \log(1 - \eta(k_*, \epsilon, \ell))$ ▷ Compute sample size t
if $n < \ell$ **then** ▷ Check if G is large enough to contain H
    $\mathcal{A}$ *rejects* G
else if $n < t$ **then** ▷ Check if G is small enough to query all edges
    query all pairs of vertices of G to $\mathcal{O}$
    if $\exists v_{i_1}, \ldots, v_{i_\ell} \in G$ such that $\{v_{i_1}, \ldots, v_{i_\ell}\}$ induces a copy of H in G **then**
        $\mathcal{A}$ *accepts* G
    else
        $\mathcal{A}$ *rejects* G
    end if
else ▷ Sample t vertices uniformly at random, without repetitions
    $S \leftarrow \emptyset$
    while $i \leq t$ **do**
        $s_i \sim G$
        while $s_i \in S$ **do** ▷ Repeat until a new vertex is sampled
            $s_i \sim G$
        end while
        $S \leftarrow S \cup \{s_i\}$
    end while
    query all pairs of vertices of S to $\mathcal{O}$
    if $\exists v_1, \ldots, v_\ell \in S$ such that $\{v_1, \ldots, v_\ell\}$ induces a copy of H in G **then**
        $\mathcal{A}$ *accepts* G
    else
        $\mathcal{A}$ *rejects* G
    end if
end if

Property Testing of H-freeness in Stable Graphs

A lower bound for the abundance η only depending on the unavoidability parameter γ, the stability k_*, and and graph H's size ℓ, is:

$$\eta \geq \left(\frac{1}{2}\right)^{\frac{\ell^3+5\ell-6}{6}} \cdot (1-\gamma)^{\frac{\ell(\ell-1)}{2}} \cdot \left(\frac{1}{24}\min\left\{\sqrt{\gamma}, \left(\frac{1}{2}\right)^{\frac{\ell(\ell-1)}{2}}(1-\gamma)^{\ell-1}\right\}\right)^{\ell\left(2^{k_*+1}-1\right)}.$$

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The resulting query complexity of the algorithm $\mathcal{A}$ can bounded by

$$q \leq \binom{t}{2} \leq \left(\frac{\ell\log(\frac{2}{3})}{\log(1-\eta_{31}(k_*, \epsilon, \ell))}\right)^2.$$

Conclusion and Future Work

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- A comparison with the Strong Regularity Lemma (for the broader class of graphs with bounded VC-dimension) reveals a potential suboptimality.
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  This raises a key question: Can the bound for the Stable Regularity Lemma be improved to match the polynomial exponent, i.e., to $(1/\epsilon)^{c\cdot k^2}$?

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- We designed an efficient *property testing algorithm* for H-freeness in stable graphs. Our algorithm leverages the most powerful feature of the Stable Regularity Lemma: its partitions have *no irregular pairs*. This provides a uniquely clean structure that is ideal for property testing.
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  This raises a key question: Can the bound for the Stable Regularity Lemma be improved to match the polynomial exponent, i.e., to $(1/\epsilon)^{c \cdot k^2}$?
- The powerful structure guaranteed by stable regularity opens up new research avenues, including:
  - Testing for forbidden subgraphs (H_n) that *grow in size* with the input graph.
  - Moving beyond freeness testing to *subgraph counting* and estimation.

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Thank you!

You can now rate the presentation. Was it *good*, *excellent* or *exceptional*?